Practice Set 17 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Various uses of intestines in Uro

Type: LAQ (25 marks) (also answered in 2023-Winter Paper 2, Q2)

Read as: the various uses of intestinal segments in urology.

Introduction

- Bowel is used to store urine, carry it to the skin, replace the ureter, make catheterisable channels and line a neovagina.
- Choice depends on renal function, previous surgery or radiation, bowel disease, mesenteric length and the reconstruction needed.

Uses at a glance

Use	Segment and operation
Incontinent diversion	**Ileal conduit** (10–15 cm ileum, 10–15 cm from the ileocaecal valve); colon conduit (transverse) after pelvic radiation; jejunal conduit only if nothing else (avoid)
Continent cutaneous pouch	Indiana (right colon, plicated ileum as channel), Mainz I, Kock (ileal nipple valves); Mitrofanoff channel
Orthotopic neobladder	Studer (40–44 cm ileum plus afferent limb), Hautmann (~70 cm ileum, W shape), joined to the urethra
Rectal reservoir	Ureterosigmoidostomy, Mainz II: now rare (acidosis, cancer)
Augmentation cystoplasty	"Clam" ileocystoplasty (20–40 cm ileum), sigmoid, caecum; gastrocystoplasty in renal failure. For refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation)
Ureteric replacement	**Ileal ureter**, isoperistaltic, for long defects; Yang–Monti tubes
Catheterisable channels	**Mitrofanoff** appendicovesicostomy; **Yang–Monti** (1–2 cm ileum retubularised transversely); MACE (Malone antegrade continence enema) for bowel
Neovagina	**Sigmoid** preferred (wide lumen, mobile mesentery, less stenosis); ileum if pelvis irradiated; MRKH (Mayer–Rokitansky–Küster–Hauser), cloaca, gender-affirming vaginoplasty

- Ileal ureter contraindications: creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease, radiation enteritis.

Choice of segment

Segment	Electrolyte disturbance	Points
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Less mucus; usable in renal failure; **haematuria–dysuria syndrome**
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis**	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Mobile, reliable supply; B12 and bile-salt loss
Colon	Hyperchloraemic metabolic acidosis	Easy antireflux tunnel; diarrhoea if ileocaecal valve used

Technical principles

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed; mark the stoma through the rectus.

- **Bowel anastomosis:** good blood supply, no tension, no spillage; close the mesenteric window.
- **Mesentery:** transilluminate, keep a major arcade; avoid watershed areas (Griffiths point, Sudeck point).
- **Detubularisation:** opening the bowel on its antimesenteric border and refolding it into a sphere **abolishes coordinated contractions** and gives more volume for the same surface area, so pressure stays low (Laplace's law).
- **Ureterointestinal anastomosis:** spatulated, tension-free, mucosa to mucosa, stented. **Wallace** (ureters joined, refluxing) has the lowest stricture rate (~3%); **Bricker** end-to-side 4–22%; antireflux tunnels (Leadbetter–Clarke) stricture more (8–14%). Reflux with normal ureters has not been shown to harm the adult kidney.
- **Stoma:** protruding "rosebud" for appliances; flush for catheterised pouches; loop-end (Turnbull) in obese patients.

Selecting conduit versus continent diversion

- **Renal function for continent diversion:** creatinine <2 mg/dL; GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria.
- **Contraindications to continent diversion:** poor renal or hepatic function, inability or unwillingness to self-catheterise, inflammatory bowel disease or short gut. A neobladder also needs a tumour-free urethra and a competent sphincter.
- A conduit is simplest and fastest; continent options improve body image but carry more reoperations and metabolic problems.

Complications

Metabolic (worse with longer segment, longer contact time, poorer renal function)

- **Hyperchloraemic acidosis:** ammonium replaces sodium on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate.
- **Hypokalaemia** (colon): replace potassium before correcting acidosis.
- **Altered sensorium:** hyperammonaemia (drain the pouch, lactulose). **Drug reabsorption:** phenytoin, methotrexate.
- **Osteomalacia**, growth delay in children; **B12 deficiency** (parenteral B12 after 5–7 years); bile-salt diarrhoea.

Other

- Bacteriuria (common; treat only symptoms or urea-splitting organisms); **stones** (reservoirs ~20%, mucus and staples); **augment perforation** (emergency; regular CIC prevents it); ureterointestinal stricture; bowel obstruction; stomal stenosis, parastomal hernia; mucus.
- **Cancer:** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years). Neovagina: stenosis, mucus discharge, adenocarcinoma reported.

Follow-up

- Lifelong electrolytes, bicarbonate, renal function, B12, upper-tract imaging, and endoscopy of colonic or augmented segments.

Recent advances

- Robotic intracorporeal diversion; tissue-engineered bladder and alternatives to bowel (ureterocystoplasty, autoaugmentation) remain limited.

Diagram to draw: ileal conduit with mesenteric arcade and Wallace anastomosis.

Key points to remember:

- Ileum and colon: hyperchloraemic acidosis; jejunum: hyperkalaemia; stomach: alkalosis.
- Continent diversion needs GFR >35, pH ≤5.8, osmolality ≥600.
- Detubularisation gives low pressure; Wallace has the lowest stricture rate.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.

Q2. Describe management of LAPC

Type: LAQ (25 marks) (related: Practice Set 10 Paper 2, Q1)

LAPC is read as locally advanced prostate cancer.

Definition and introduction

- **cT3–4, N0 or N1, M0:** extracapsular extension (T3a), seminal vesicle invasion (T3b), fixed or invading adjacent organs (T4), ± pelvic nodes.
- At least 10% of new cases. Any single treatment carries a high recurrence risk, so **multimodal therapy** is the rule. In the PSA (prostate-specific antigen) era, high risk is defined more by grade and PSA than by palpation.
- Untreated, 15-year disease-specific survival is 57% for T3–T4 vs 81% for T1–T2 (Swedish cohort).

Evaluation

- **DRE (digital rectal examination)**, PSA, biopsy ISUP (International Society of Urological Pathology) grade group.
- **mpMRI (multiparametric MRI):** separates T2 from T3 (PI-RADS v2 reporting).
- **Bone scan + CT/MRI**, or **PSMA (prostate-specific membrane antigen) PET-CT**. proPSMA trial: accuracy 92% vs 65% for conventional imaging.
- NCCN (National Comprehensive Cancer Network) very high risk: cT3b–T4, primary Gleason pattern 5, or >4 cores grade group 4–5. **Also know:** CAPRA (Cancer of the Prostate Risk Assessment) score; Decipher.
- Assess life expectancy: a fit man in his 70s with high-grade disease will live long enough to die of it.

A. Radiotherapy (RT) + long-term ADT (androgen deprivation therapy): standard

- Dose-escalated IMRT (intensity-modulated RT, ≥78–81 Gy), or EBRT (external-beam RT) with brachytherapy boost (**ASCENDE-RT:** LDR (low-dose-rate) boost halved biochemical failure; overall survival (OS) equal).
- **Why add ADT:** it shrinks the target and is synergistic with radiation.
- **EORTC 22863:** 3 years of goserelin with RT: 5-year OS 78–79% vs 62%.
- **RTOG 92-02:** 28 vs 4 months ADT better on all endpoints except overall OS; OS gain in Gleason 8–10.
- **Duration:** NCCN 18–36 months with EBRT (12–36 with brachytherapy boost); 18 vs 36 months gave similar OS.
- **RT is needed with ADT:** SPCG-7 (cancer death RR (relative risk) 0.44) and MRC PR07 (OS HR (hazard ratio) 0.77).
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT favoured it in high-risk N0.

B. Systemic intensification

- **STAMPEDE (M0):** for N1, or two of T3–4, Gleason ≥8, PSA ≥40, **abiraterone + prednisolone for 2 years** with ADT + RT: metastasis-free survival HR 0.53, OS HR 0.60. NCCN lists it for very high risk.
- Docetaxel (RTOG 0521 OS HR 0.69): not routine.

C. Radical prostatectomy (RP) + ePLND (extended pelvic lymph node dissection)

- For fit men with a resectable tumour, as part of multimodal therapy. cT3 is overstaged in 7–27%. RP alone gives disease-free survival in at least half at 8–10 years, and 10-year disease-specific survival ~80–90%.
- **Advantages:** accurate pathological staging; removes the source of local morbidity.
- **Neoadjuvant ADT before RP:** clinical downstaging and fewer positive margins, but **no survival gain**.
- **Post-RP RT:** EORTC 22911, SWOG 8794 (metastasis-free survival and OS improved) and ARO 96-02 favour adjuvant RT over observation. **RADICALS-RT, RAVES, GETUG-AFU 17: early salvage RT** is preferred for most men. Adjuvant RT is still considered for very high risk (pT3b, Gleason 8–10, positive margin). Dose 64–66 Gy; adding short-term ADT to salvage RT improved progression-free survival (GETUG-AFU 16).

D. ADT alone

- Only for men unfit for or refusing local treatment. Consider immediate ADT for PSA doubling time <12 months, PSA >50 ng/mL or symptoms; otherwise deferred ADT is acceptable.
- **Also know:** bicalutamide 150 mg (spares sexual interest; harmful in localised disease); intermittent ADT.

Node-positive disease

Setting	Management
cN1 M0	RT to prostate and pelvis + ADT 2–3 years + abiraterone 2 years; RP + ePLND in selected men
pN1 after RP	**Immediate ADT** (ECOG 7887: OS benefit) **or ADT + adjuvant RT** (better survival than ADT alone); observation if PSA undetectable and ≤2 nodes

Algorithm

```
cT3–4 / cN1, M0 (PSMA PET or CT + bone scan)
   ├─ Fit, resectable  → RP + ePLND → early salvage RT ± ADT
   ├─ Most             → RT + ADT 18–36 months (+ abiraterone if N1 / very high risk)
   └─ Unfit / refuses  → ADT alone (immediate or deferred)
```

Complications and local sequelae

- ADT: hot flushes, osteoporosis, metabolic syndrome, cardiovascular risk, loss of libido; give bone protection.
- RT: proctitis, cystitis. RP: incontinence, erectile dysfunction.
- A prostate left in place can cause outflow obstruction, haematuria and ureteric obstruction.
- **Also know:** cryotherapy and HIFU (high-intensity focused ultrasound) are not standard.

Follow-up and recent advances

- PSA every 3–6 months; bone and metabolic health.
- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

Diagram to draw: T-stage diagram (T3a extracapsular, T3b seminal vesicle, T4 adjacent organs).

Key points to remember:

- RT + long-term ADT (18–36 months) is standard; ADT alone is inferior (SPCG-7).
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0.
- Neoadjuvant ADT before RP: no survival gain.
- Early salvage RT preferred over adjuvant RT for most men after RP.

Q3a. Drug therapy in OAB

Type: SAQ (10 marks) (also answered in Practice Set 1 Paper 1, Q3e)

Place and mechanisms

- In OAB (overactive bladder), drugs are second line, after behavioural therapy. Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.
- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent signalling.
- **β3 agonists:** activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage.

Agents

Drug	Feature	Usual dose
Oxybutynin	Non-selective; most dry mouth; patch, gel	5 mg two to three times daily; ER (extended-release)
Tolterodine	Bladder-selective; QT prolongation at high dose	ER 4 mg daily
Fesoterodine	Prodrug of tolterodine's active metabolite	4–8 mg daily
Solifenacin	M3-selective; long half-life	5–10 mg daily
Darifenacin	About 30-fold M3-selective	7.5–15 mg daily
Trospium	Quaternary; little brain entry	20 mg twice daily
Mirabegron (β3)	Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise	25–50 mg daily
Vibegron (β3)	No CYP2D6 interaction	75 mg daily

Comparison

Feature	Antimuscarinics	β3 agonists
Efficacy	Similar	Similar
Side effects	Dry mouth, constipation, blurred vision	Near placebo
Cognition	**Dementia risk 10–65% higher with use >3 months**	No anticholinergic burden
Contraindications	Retention, uncontrolled narrow-angle glaucoma, gastric retention	Uncontrolled severe hypertension

- **Choice:** elderly → β3 agonist, or ER trospium, darifenacin or fesoterodine. ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.

Combination and special groups

- **Solifenacin + mirabegron** beats monotherapy (SYNERGY; BESIDE in non-responders).
- **Men:** add an antimuscarinic or β3 agonist to an α-blocker; retention up to 3%, so check PVR (post-void residual).
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; third line onabotulinumtoxinA 100 U and neuromodulation.

Diagram to draw: detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.

Key points to remember:

- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk.
- Solifenacin + mirabegron combination works (SYNERGY).

Q3b. Techniques of chordee correction

Type: SAQ (10 marks) (related: Practice Set 10 Paper 1, Q3d)

Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test
  ├─ Straight        → urethroplasty
  ├─ Residual <30°   → dorsal plication
  └─ >30°            → divide urethral plate → still >30° → ventral lengthening (stage)
```

	Dorsal shortening (plication)	Ventral lengthening (corporotomy ± graft)
Methods	**Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication	1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect
Use	≤15–30°	Moderate–severe after plate division
Recurrence	28% (one series)	9%
Risk	Shortening; dorsal nerve injury off midline	Longer, staged; theoretical erectile dysfunction

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

Diagram to draw: penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

Q3c. Anterior urethral valve

Type: SAQ (10 marks)

Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

AUV versus PUV

Feature	AUV	PUV
Frequency	25–30 times rarer	Commonest infant outlet obstruction
Site	Bulbar, penoscrotal, penile	From the verumontanum
Sign	Penile bulge	Weak stream, palpable bladder
Bladder dysfunction	Up to 80%	Near universal
Renal outcome	78% normal function	ESRD (end-stage renal disease) 20–50%

Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

Diagram to draw: VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

Key points to remember:

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function.

Q3d. Vaginal prolapse repair.

Type: SAQ (10 marks) (related: Practice Set 12 Paper 1, Q3d)

Read as: surgical repair of vaginal wall and vault prolapse (pelvic organ prolapse).

Principles

- POP (pelvic organ prolapse) is a quality-of-life condition: operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q (Pelvic Organ Prolapse Quantification). **The apex is the cornerstone**: failing to support it greatly raises recurrence.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI (stress urinary incontinence)** with prolapse reduced; consider a concomitant sling. Finish with **cystoscopy** to confirm ureteric efflux.

Operations by compartment

Compartment	Operations	Key points
Anterior (cystocele)	**Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP (arcus tendineus fasciae pelvis))	Central vs lateral defect; does not support the apex
Apical (uterus, vault)	**Uterosacral suspension** (ureter injury 1–11%); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; hysteropexy; Manchester	OPTIMAL trial: uterosacral = sacrospinous at 2 years
Apical, abdominal	**Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to sacral promontory	Most durable (>90%); mesh exposure ~10% long term (CARE)
Posterior (rectocele)	**Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy	Levator plication causes dyspareunia; constipation may persist
Obliterative	**Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis	Elderly, not sexually active; success >90%; coital regret ≤13%

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- Transvaginal synthetic mesh for POP was withdrawn in the USA (FDA (Food and Drug Administration), 2019).

Complications

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia; recurrence (anterior after sacrospinous fixation); mesh exposure.

Diagram to draw: DeLancey levels of support with sacrospinous and sacrocolpopexy fixation points.

> **Key points to remember:**
> - Always restore apical support.
> - Sacrocolpopexy most durable; colpocleisis for frail elderly.
> - Cystoscopy after repair; screen for occult SUI.

Q3e. Perineo-scrotal hypospadias

Type: SAQ (10 marks) (related: 2017-Winter Paper 3, Q1)

Definition

- A severe **proximal hypospadias**: meatus at the penoscrotal junction, scrotum or perineum, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

Evaluation

- Examine stretched penile length, glans width, transposition and **both testes**.
- **DSD (disorder of sex development) work-up** if there is an undescended testis (especially non-palpable), a perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone). DSD risk is 64% in proximal vs 7% in distal cases.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

Preparation

- **Timing:** 6–18 months.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery; evidence mixed.

Surgery

1. Curvature correction: deglove, artificial erection, plication if <30°, plate division and ventral corporotomy ± graft if >30° (see Practice Set 17 Paper 2, Q3b).

2. Urethroplasty: staged repair is now preferred

Repair	Technique	Comment
Two-stage Bracka / STAG (staged tubularised autograft)	Stage 1: correct curvature, divide plate, quilt inner-preputial or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer	Preferred for most
Two-stage Byars	Dorsal preputial flaps moved ventrally, tubularised later	Supple flap bed
One-stage TPIF (transverse preputial island flap, Duckett tube) or onlay island flap	Flap urethroplasty	Higher complications
Proximal TIP (tubularised incised plate)	Plate kept	Stricture, recurrent curvature: abandoned for severe cases

- **Why stage?** A urethroplasty over a fresh corporal graft fails more often.
- Principles: magnification, tension-free closure, **vascularised barrier layer** (dartos, tunica vaginalis); **scrotoplasty** for transposition.

Complications

- **23–68%**; 25–75% need further surgery. **Fistula** commonest (3–45%), glans dehiscence, meatal stenosis, stricture, **diverticulum** (long rigid neourethra raises resistance: Poiseuille's law), recurrent curvature, hair-bearing urethra.

Diagram to draw: two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

Key points to remember:

- Perineoscrotal hypospadias with undescended testis → DSD work-up.
- Curvature >30°: plate division ± graft and staging.
- STAG/Bracka staged repair preferred; fistula commonest complication.

Q3f. Surgical management of bilateral varicocele

Type: SAQ (10 marks) (related: Practice Set 6 Paper 3, Q3e)

Introduction

- Dilated pampiniform veins: ~15% of men, 35% of primary and 75–81% of secondary infertility. Mostly left (left vein enters the renal vein 8–10 cm higher). With sensitive tests most are bilateral; clinical significance of subclinical ones is uncertain.
- Harm: heat (lost countercurrent exchange), oxidative stress, DNA fragmentation, Leydig dysfunction; progressive.
- Grades: subclinical (Doppler only), I (palpable on Valsalva), II (palpable), III (visible).

Indications for repair

- Infertile couple, **palpable** varicocele, abnormal semen, normal female partner.
- Pain (dull ache on standing), hypogonadism, high DNA fragmentation; adolescent testicular asymmetry >20%.
- **Do not repair subclinical** sides. Repair each palpable side.

Techniques

Technique	Recurrence	Hydrocele	Bilateral use
Microsurgical subinguinal / inguinal	**0.5–1%**	**~0**	**Preferred**: two small incisions, local or regional anaesthesia
Laparoscopic	3–15%	12%	Both sides through same ports; general anaesthesia, serious complications possible
Retroperitoneal (Palomo)	15–25%	7%	Children; artery ligation acceptable
Embolisation / sclerotherapy	15–25%	0	Femoral (left) or jugular (right) access; technical failure

Microsurgical steps (each side)

- Subinguinal incision over the external ring; deliver the cord.
- Microscope and **micro-Doppler** to map the testicular artery; papaverine for spasm.
- Ligate all internal and external spermatic veins; preserve **artery, lymphatics, vas and one set of vasal veins**.
- Deliver the testis to ligate gubernacular veins (lower recurrence).
- **Impulse test** confirms no missed vein.

Bilateral-specific points

- **Never ligate both testicular arteries**: in adults this has caused azoospermia and atrophy. So artery-sparing microsurgery is safest bilaterally.
- An isolated right varicocele, or one that does not decompress lying down: image for a renal or retroperitoneal mass.
- Repair any coexisting hydrocele.

Complications

- Hydrocele (lymphatic ligation), testicular artery injury (atrophy), recurrence (upright colour Doppler shows reflux).

Diagram to draw: spermatic cord cross-section: artery, lymphatics and vas preserved, veins ligated.

> **Key points to remember:**
> - Repair only palpable sides with abnormal semen.
> - Microsurgical subinguinal: recurrence <1%, hydrocele nil.
> - Never ligate both testicular arteries.

Rapid Revision Sheet – Practice Set 17 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Various uses of intestines in Uro (LAQ)

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- Continent diversion needs GFR >35, pH ≤5.8, osmolality ≥600.
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- RT + long-term ADT (18–36 months) is standard; ADT alone is inferior (SPCG-7).
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0.
- Neoadjuvant ADT before RP: no survival gain.
- Early salvage RT preferred over adjuvant RT for most men after RP.

Q3a. Drug therapy in OAB (SAQ)

- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk.
- Solifenacin + mirabegron combination works (SYNERGY).

Q3b. Techniques of chordee correction (SAQ)

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

Q3c. Anterior urethral valve (SAQ)

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function.

Q3d. Vaginal prolapse repair. (SAQ)

- Always restore apical support.
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.

Q3e. Perineo-scrotal hypospadias (SAQ)

- Perineoscrotal hypospadias with undescended testis → DSD work-up.
- Curvature >30°: plate division ± graft and staging.
- STAG/Bracka staged repair preferred; fistula commonest complication.

Q3f. Surgical management of bilateral varicocele (SAQ)

- Repair only palpable sides with abnormal semen.
- Microsurgical subinguinal: recurrence <1%, hydrocele nil.
- Never ligate both testicular arteries.